Criterion C - System Overview

System Models

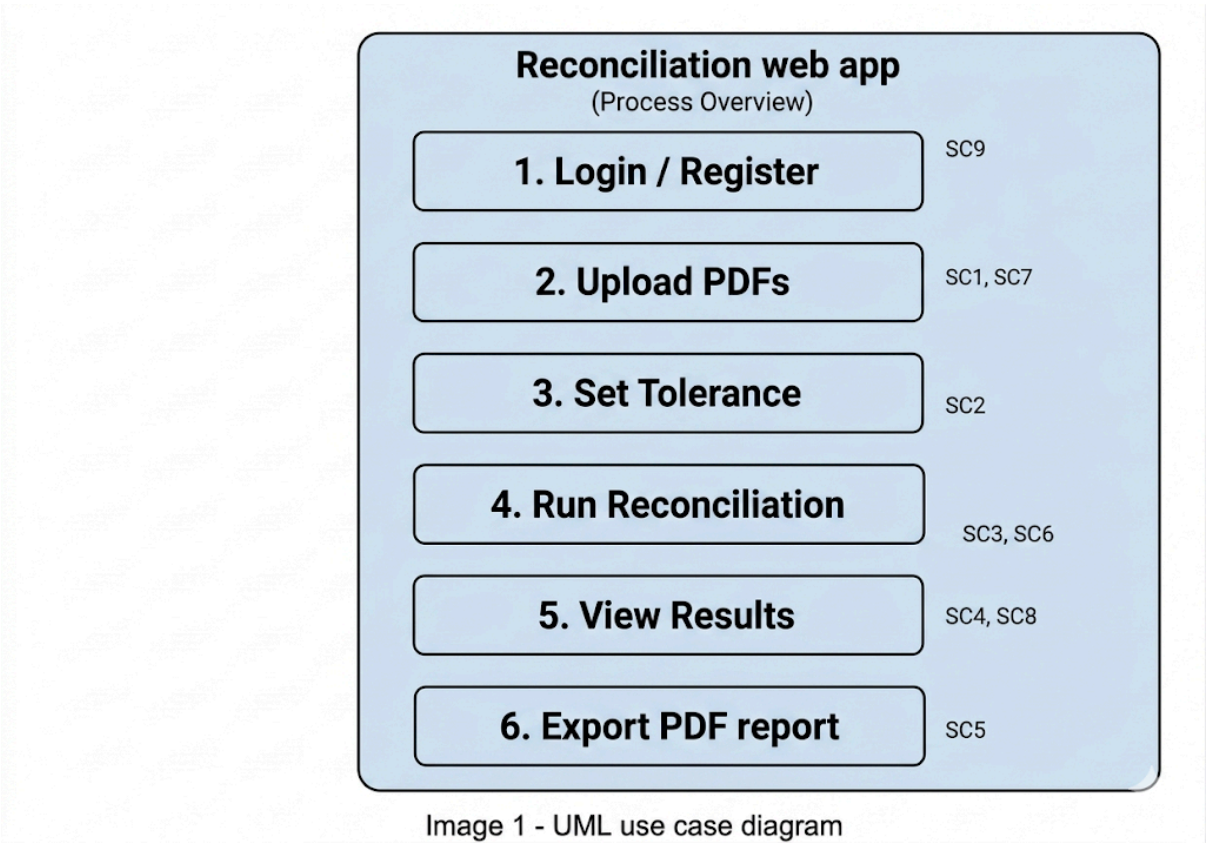


Image 1 - UML use case diagram

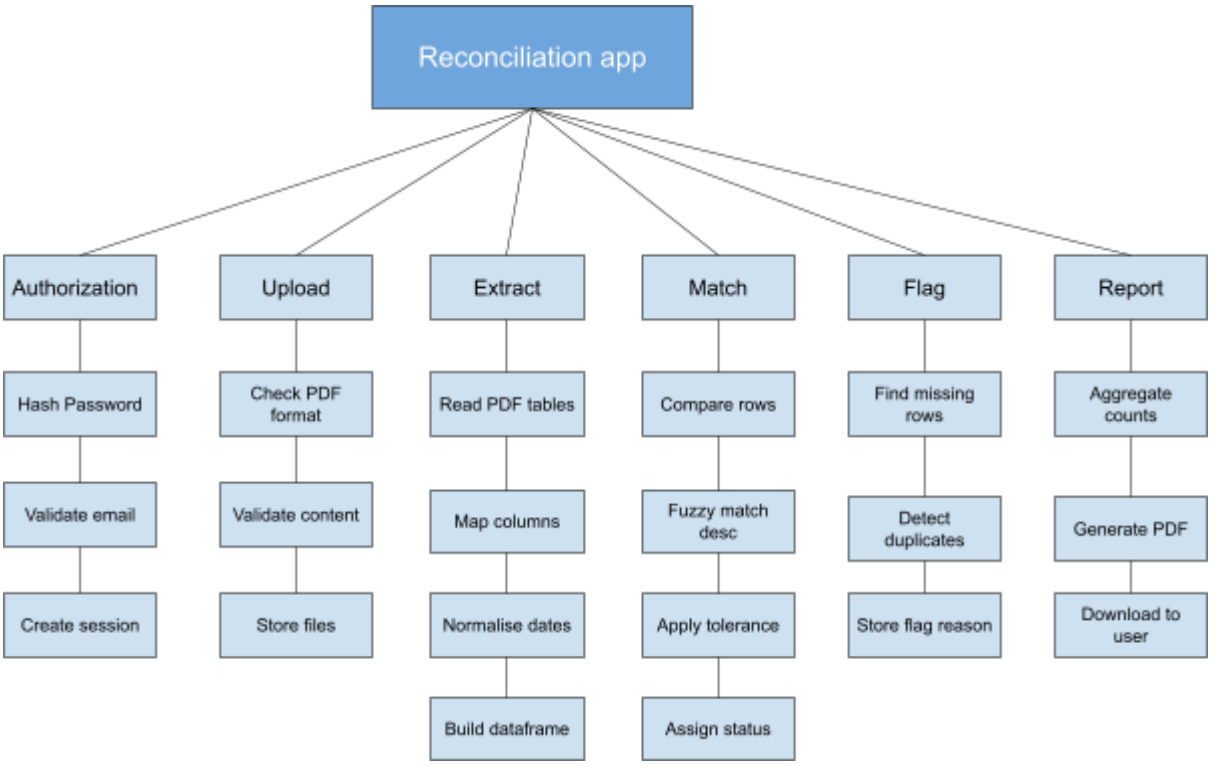


Image 2 - Hierarchy Chart

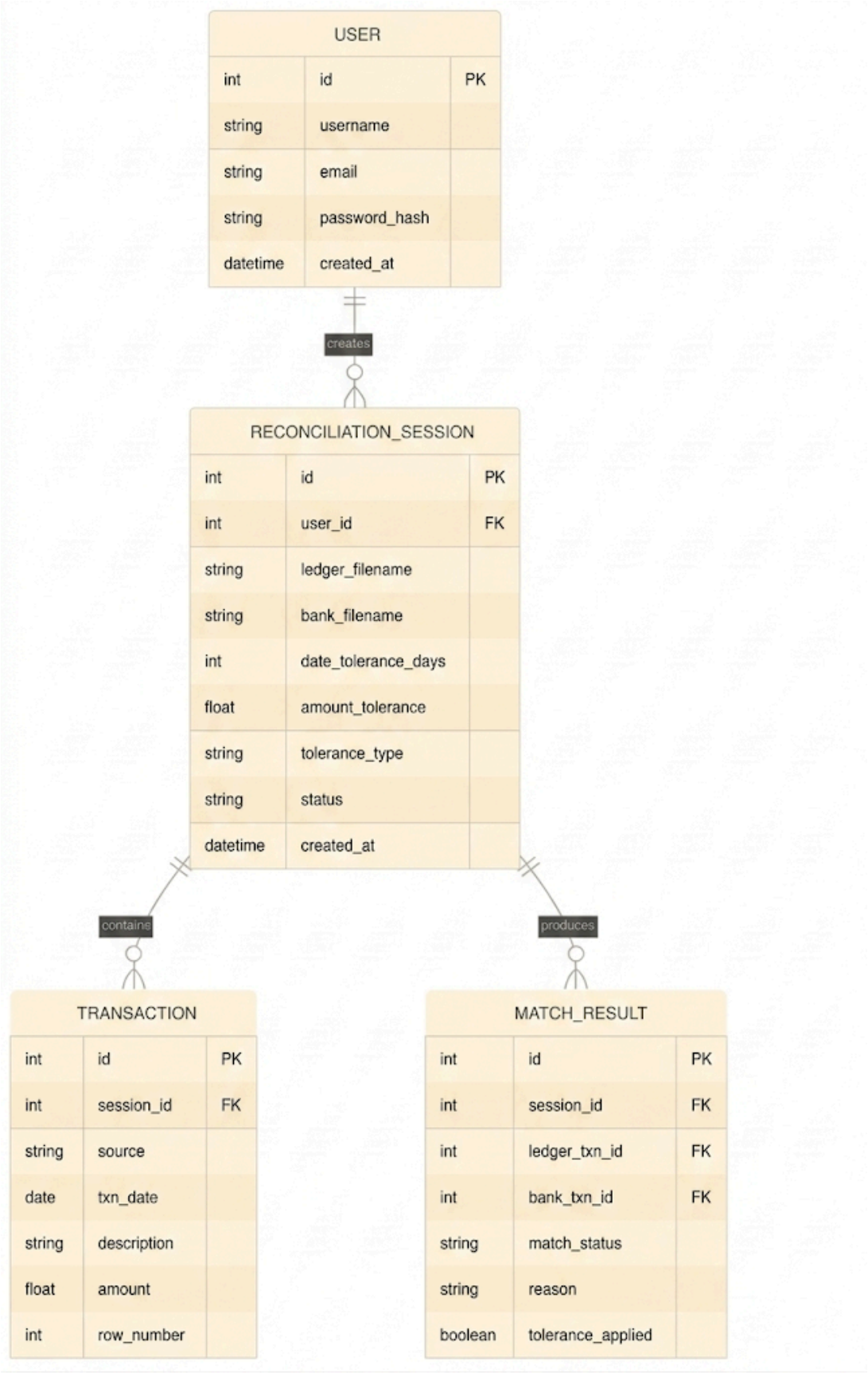


Image 3 - ERD diagram

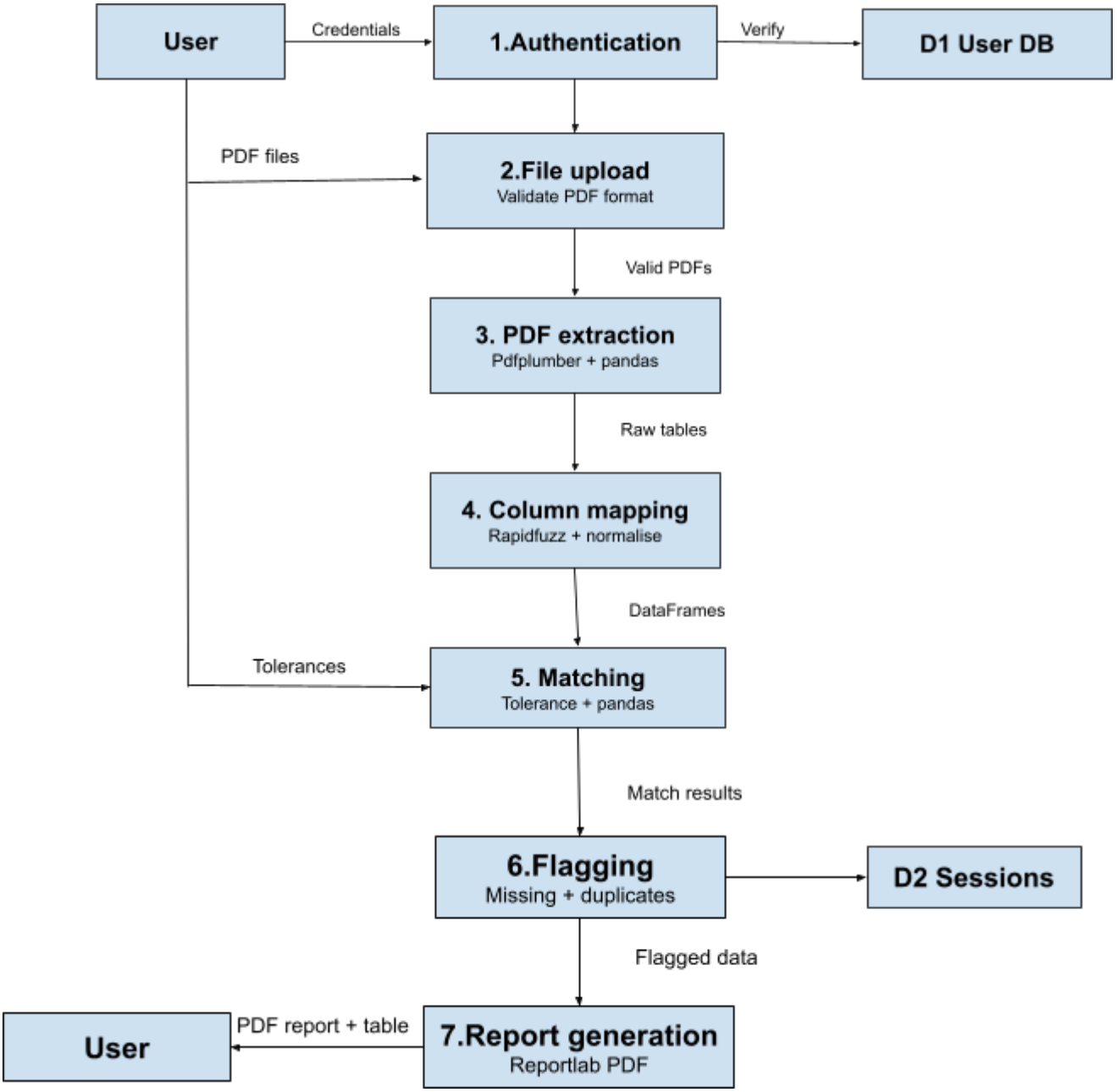
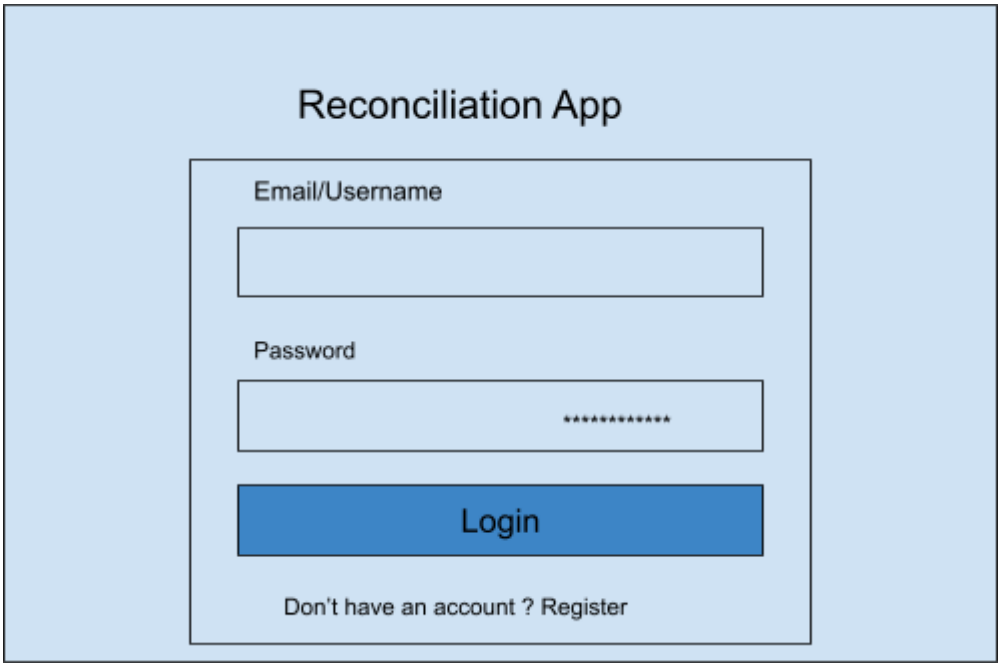

User
Credentials
1.Authentication
Verify
D1 User DB
PDF files
2.File upload
Validate PDF format
Valid PDFs
3. PDF extraction
Pdfplumber + pandas
Raw tables
4. Column mapping
Rapidfuzz + normalise
DataFrames
Tolerances
5. Matching
Tolerance + pandas
Match results
6.Flagging
Missing + duplicates
D2 Sessions
Flagged data
User
PDF report + table
7.Report generation
Reportlab PDF

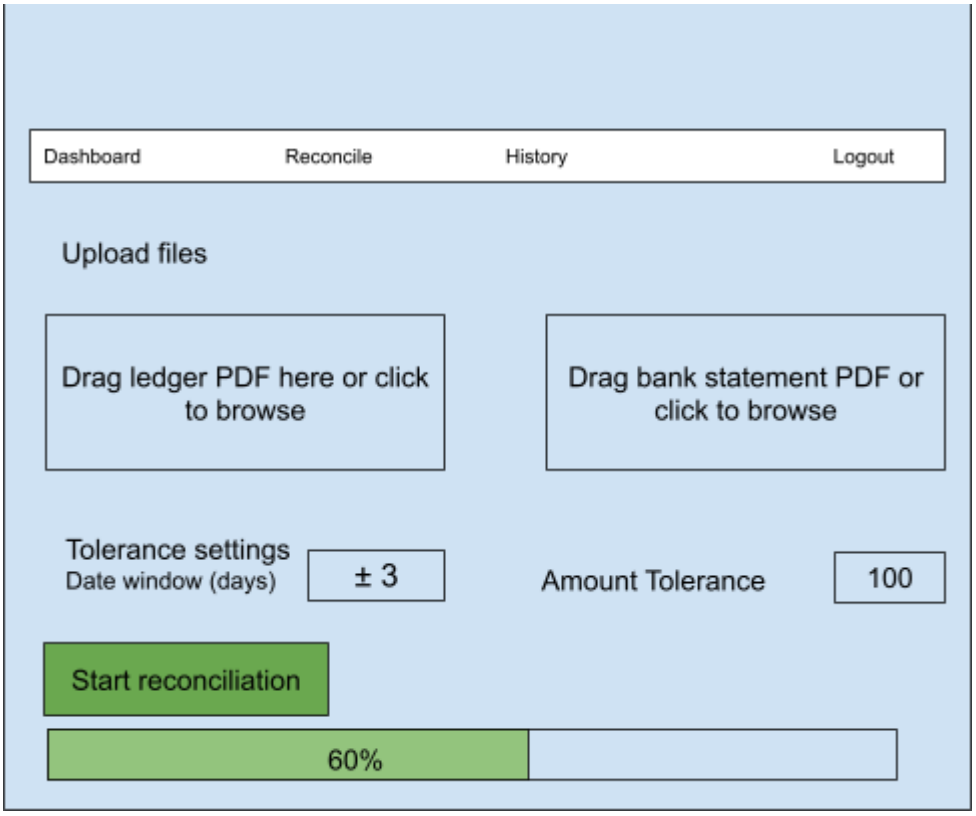
Image 4 - DFD

GUI

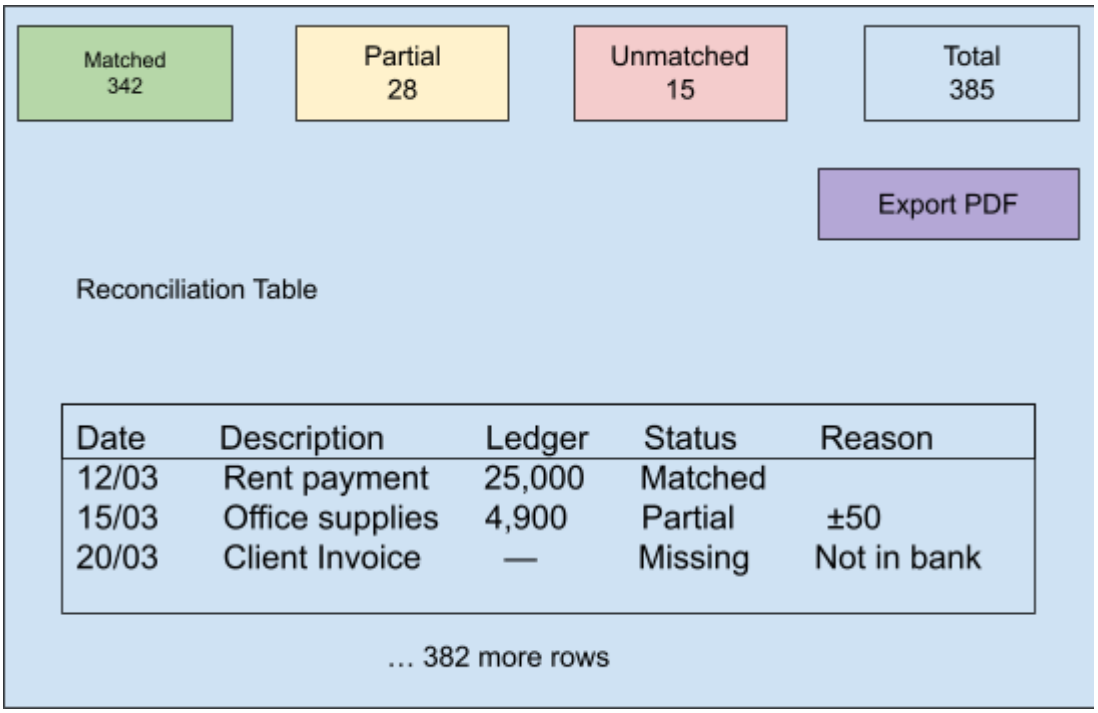
Screen 1 - Login page

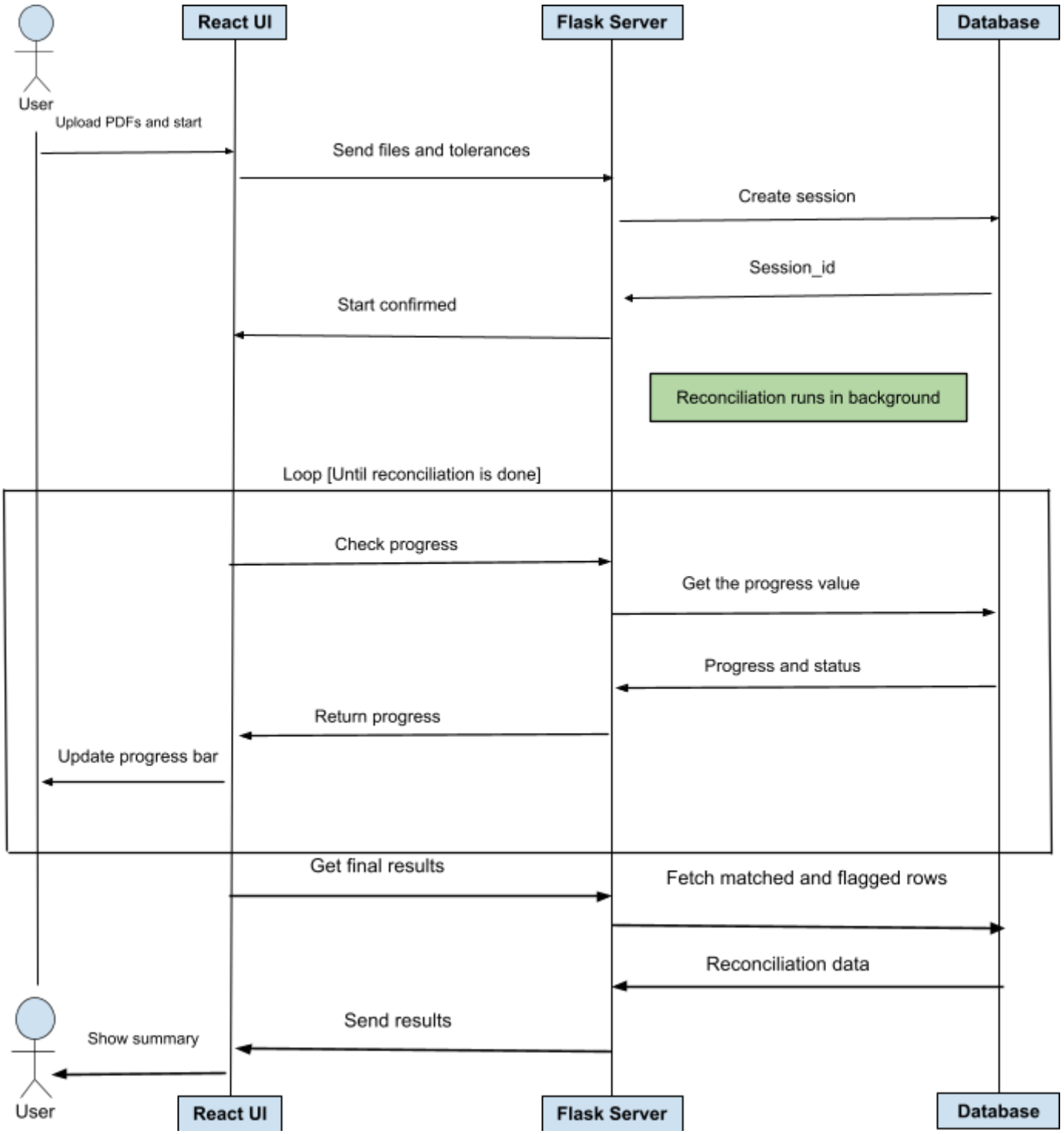


Screen 2 - Upload and Tolerance settings



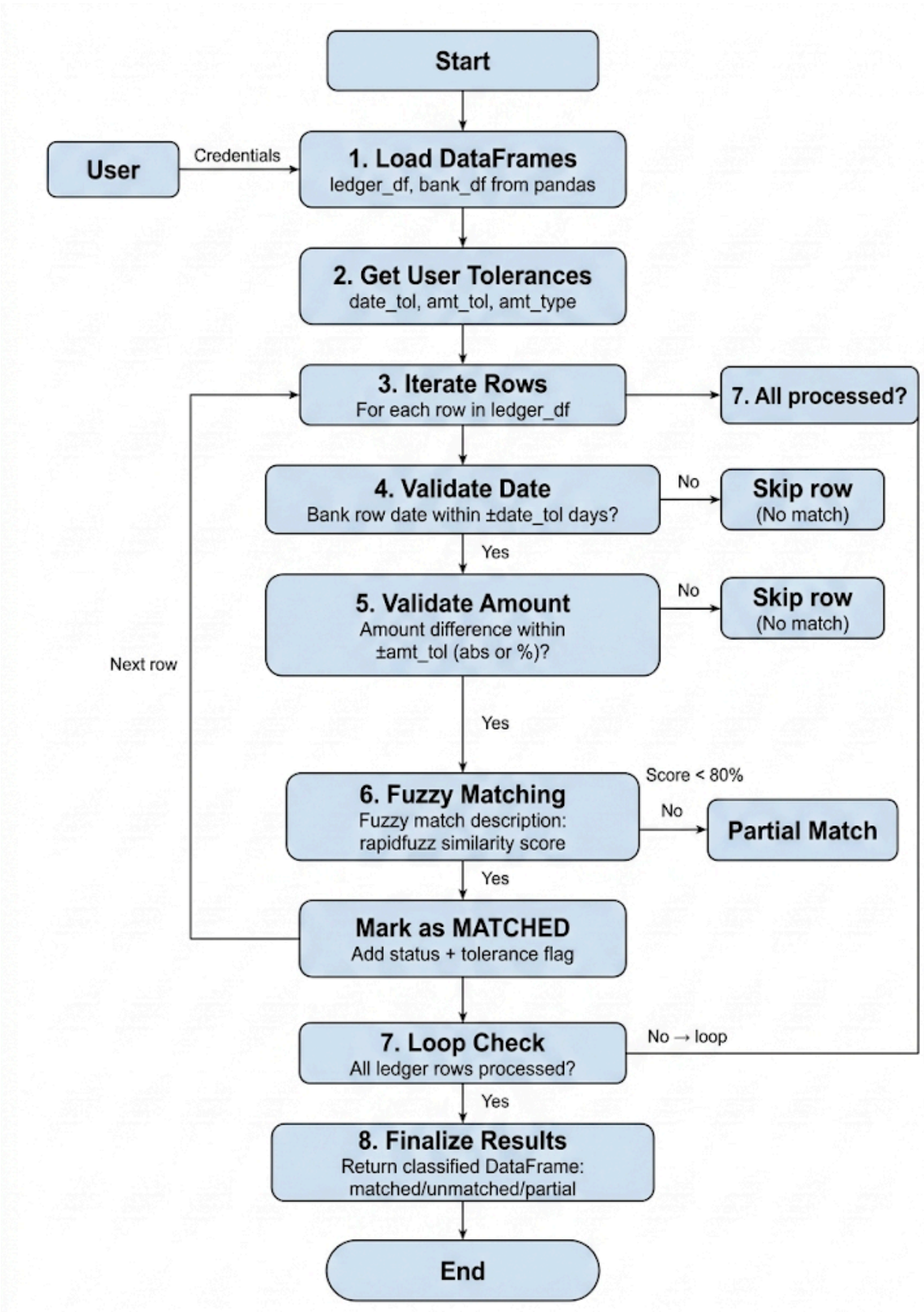
Screen 3 - Reconciliation results



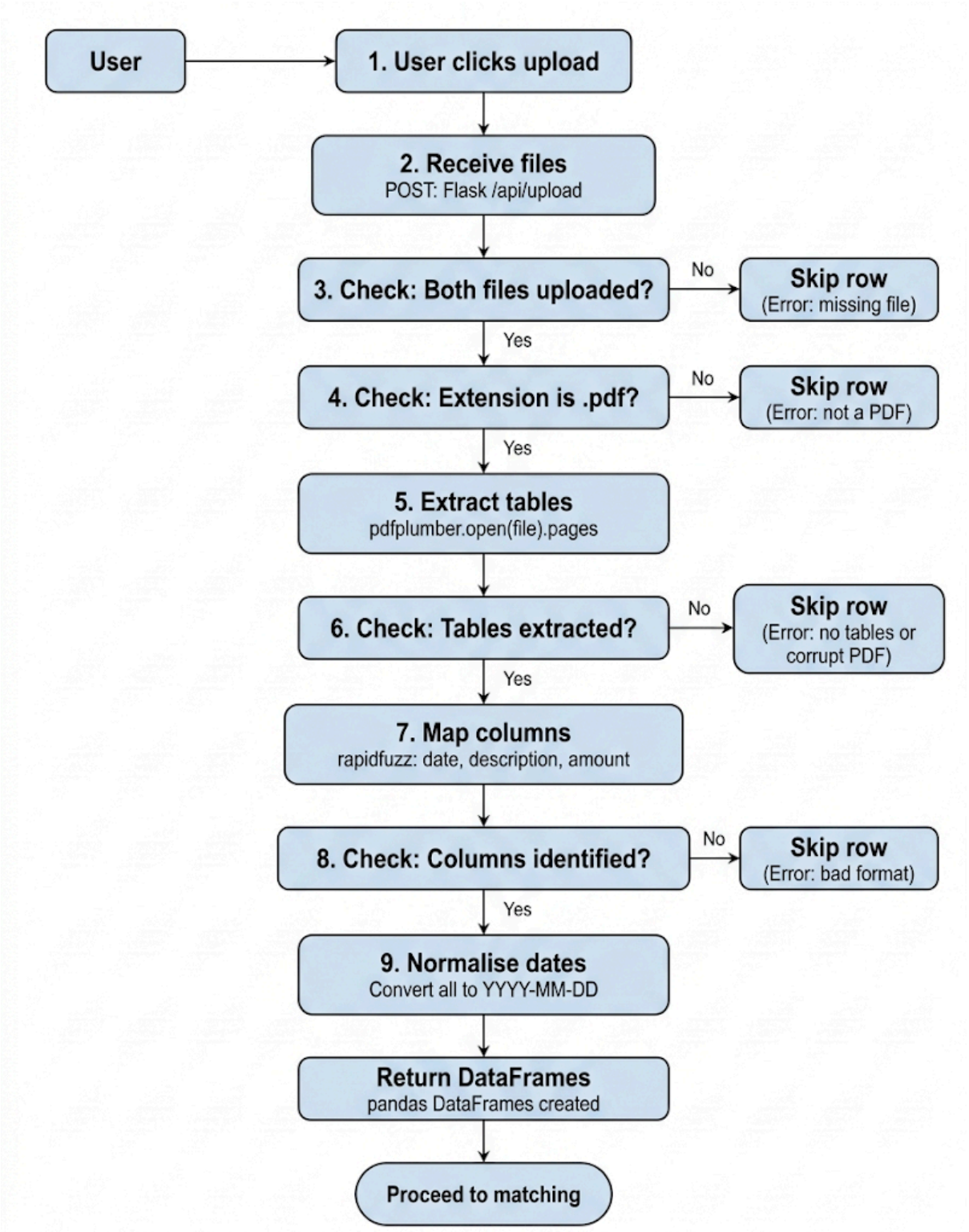


sequence diagram

Algorithm



Transaction Matching Flowchart



File Upload Validation Flowchart

Pseudocode

1. Flagging the unmatched and irregular transaction

FUNCTION flag_transactions(ledger_df, bank_df, match_results):
flagged = empty list

// 1. Flag rows present in ledger but missing from bank
FOR each row in ledger_df:IF row.id NOT IN match_results.ledger_ids:ADD row to flagged with reason "Missing from bank"

// 2. Flag rows present in bank but missing from ledger
FOR each row in bank_df:IF row.id NOT IN match_results.bank_ids:ADD row to flagged with reason "Missing from ledger"

// 3. Detect duplicates within each file
FOR each df in [ledger_df, bank_df]:duplicates = df where (date, description, amount) appears > 1FOR each dup in duplicates:ADD dup to flagged with reason "Duplicate record"

// 4. Flag tolerance-applied matches
FOR each match in match_results:IF match.tolerance_applied == True:ADD match to flagged with reason "Matched with tolerance"

RETURN flagged

2. User Authentication with password hashing

FUNCTION register(username, email, password):
IF username or email already exists in database:
RETURN error "Account already exists"
hashed = werkzeug.generate_password_hash(password)
INSERT new User(username, email, hashed) into database
RETURN success

FUNCTION login(username, password):
user = query database WHERE username matches
IF user NOT found OR check_password_hash fails:
RETURN error "Invalid credentials"
SET session cookie with user.id
RETURN redirect to /reconcile

Testing Strategy

Success Criteria	Scenerio	Test data	Testing Strategy	Expected Result
File Upload	Normal	Upload valid text based ledger and bank statement in pdf format	Try uploading valid format	File accepted and reconciliation begins
File Upload	Abnormal	Upload a .docx file instead of .pdf file	Try uploading invalid format	Error message displayed
File Upload	Abnormal	Upload a password-protected pdf	Try uploading an encrypted pdf	Error message displayed
File Upload	Abnormal	Upload a corrupted pdf	Try uploading a broken pdf	Error message displayed
File Upload	Abnormal	Upload empty 0-byte file	Try uploading empty file	Error message displayed
Tolerance	Normal	Set a date window and amount	Upload files with tolerance set to 2 days and amount 50	Transaction matched with tolerance flag
Tolerance	Abnormal	Set date window and amount to 0	Upload files with tolerance set to 0 days and 0 amount	Only exact matches succeed
Automatic Matching	Normal	Two identical transactions	Upload matching ledger and bank files	All transactions being shown as matched
Flagging	Normal	Transaction in ledger but not in bank	Upload field with missing entry	Missing entity being flagged
Flagging	Normal	Duplicate entry in ledger	Upload file with repeated rows	Duplicated rows being flagged
Performance	Normal	500 rows per file	Upload standard files	Reconciliation done in 10-20 s
Performance	Abnormal	10,000 rows per file	Upload bulk data files	Error message displayed
Report	Normal	Click export after reconciliation	Use the export PDF button	PDF downloads

Security	Abnormal	Try accessing without login	Reconcile without session	Redirected to login page
UI feedback	Normal	Start reconciliation	Watch progress bar	Progress bar fills during processing